

The empirical recurrence for OEIS A317700, recovered from its tail

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8 September 2026

Abstract

OEIS A317700 records “Empirical recurrence of order 87” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 109$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 91$. Its last coefficient is $c_{87} = -1747968 \neq 0$, so no further tail satisfies anything shorter; any order-87 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A317700 is “Number of nX5 0..1 arrays with every element unequal to 0, 2, 3, 4, 5, 6, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 82, 851, 20235, 468002, 11013057, 261020405, 6206695998, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Aug 04 2018 20:17:23, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 109$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 109$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 87: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 87.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 218$ exact terms from $n = 4$ on, it returns order 87 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	26	c_{23}	−3867238939028	c_{45}	−44277318053765327	c_{67}	954452387975834
c_2	−23	c_{24}	−1362435302355	c_{46}	−23710456026421611	c_{68}	382772099844778
c_3	−124	c_{25}	−7979685577820	c_{47}	20444947540726230	c_{69}	−343439765719964
c_4	−12820	c_{26}	−47790743989421	c_{48}	30529517880450233	c_{70}	−311507944392341
c_5	−22661	c_{27}	−97555305168630	c_{49}	4853042093410924	c_{71}	2536499007268
c_6	110905	c_{28}	19561963741989	c_{50}	−15078855793441992	c_{72}	108438159994319
c_7	1667865	c_{29}	473652564163533	c_{51}	−4778035537823172	c_{73}	39666800122034
c_8	4827393	c_{30}	937353905551023	c_{52}	3366600119101547	c_{74}	−14969165155020
c_9	−8545485	c_{31}	403771242411793	c_{53}	−10191438440361368	c_{75}	−14158254924744
c_{10}	−105238575	c_{32}	−1980485836064904	c_{54}	−21991562299153510	c_{76}	−295228467578
c_{11}	−353573858	c_{33}	−4621637429207616	c_{55}	−1517511229090261	c_{77}	2359577563704
c_{12}	123284093	c_{34}	−3366686832809087	c_{56}	28087263353216386	c_{78}	504809179876
c_{13}	3522342537	c_{35}	4574086309847788	c_{57}	28687597760966559	c_{79}	−292802976376
c_{14}	11927727186	c_{36}	14193907033417675	c_{58}	1761631451835257	c_{80}	−89617820836
c_{15}	6379783515	c_{37}	12773884530797142	c_{59}	−19894155517362706	c_{81}	19073188792
c_{16}	−61770907490	c_{38}	−6348642415225237	c_{60}	−16438465295037992	c_{82}	8667330016
c_{17}	−201220213092	c_{39}	−29056106739816278	c_{61}	−1116597142326457	c_{83}	283535552
c_{18}	−199557672911	c_{40}	−27637314113724750	c_{62}	6943526928740820	c_{84}	−699333696
c_{19}	503855901286	c_{41}	6402504601064891	c_{63}	4096716340694779	c_{85}	−1163648
c_{20}	1689513806460	c_{42}	42336377953514614	c_{64}	−555627751473943	c_{86}	37677056
c_{21}	1752328576671	c_{43}	34958956183721188	c_{65}	−1758882564746631	c_{87}	−1747968
c_{22}	−1042761164093	c_{44}	−11142961248355933	c_{66}	−82259474426941		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 91$, and it is the recurrence OEIS A317700 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{87} - c_1x^{86} - \dots - c_{87}$. Its constant term is $-c_{87} = 1747968$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 87 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 87, so $\deg q = 87$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 87, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 87, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1747968 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-87 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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